

PAPER_408 — Resonance MUGE Complete 14-Term Sum with Wormhole as 14th Term

PAPER_395 (Session 107) extracted the standalone wormhole acceleration formula a_{worm} .

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Source: groksharecfdcad2f5.txt, lines 277–1600 ("Star Magicconstruction file04Oct2025.docx" C++ implementation) **Section:** C++ source — compute_resonance_MUGE() function with compute_a_wormhole() as 14th additive term **Session:** 108 (groksharecfdcad2f5.txt construction file re-analysis) **CP4 Class:** ResonanceMUGE14TermCompleteWormholeSumCalculator (#57)

1. Overview

PAPER371 (Session 101) established the **12-term MUGE Superconductive Resonance** co-sum. PAPER395 (Session 107) extracted the standalone wormhole acceleration formula a_{worm} .

PAPER_408 establishes the **complete 14-term resonance MUGE**, where the wormhole term is the **14th additive component** of the resonance sum — not a standalone formula but an **integrated resonance MUGE term**:

$$g_{\text{res},14} = \underbrace{a_{\text{DPM}} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super}} + a_{\text{aether}} + U_{\text{g4i}}}_{\text{Terms 1–6}} + \underbrace{a_{\text{quantum}} + a_{\text{Aether}} + a_{\text{fluid}} + a_{\text{osc}} + a_{\text{exp}} + f_{\text{TRZ}}}_{\text{Terms 7–12}} + \underbrace{a_{\text{worm}}}_{\text{Term 14}}$$

Note: Term 13 = $f_{\text{TRZ}} = 0.1$ and Term 14 = a_{worm} as confirmed by the construction file compute_resonance_MUGE() implementation.

2. Complete 14-Term Formula

2.1 All 14 Terms

#	Term	Formula
1	a_{DPM}	$F_{\text{DPM}} / M = E_{\text{vac}} \cdot f_{\text{DPM}} \cdot V_{\text{sys}} \cdot a_{\text{DPM,base}} / (c \cdot E_{\text{vac,ISM}})$
2	a_{THz}	$10 \cdot f_{\text{THz}} \cdot v_{\text{exp}} / c \cdot a_{\text{DPM}}$
3	$a_{\text{vac,diff}}$	$(E_0 \cdot f_{\text{vac}} \cdot V_{\text{sys}} \cdot a_{\text{DPM}}) / \hbar$

#	Term	Formula
4	a_{super}	$A_{\text{sc}} \cdot a_{\text{DPM}}$
5	$a_{\text{aether, res}}$	$f_{\text{aether}} \cdot E_{\text{vac, neb}} \cdot V_{\text{sys}} \cdot a_{\text{DPM}} / (E_{\text{vac, ISM}} \cdot c)$
6	U_{g4i}	$k_4 \cdot \rho_v \cdot M_{\text{bh}} / d_g$ (BH vacuum coupling)
7	a_{quantum}	$10 \cdot f_q \cdot E_{\text{vac, neb}} \cdot V_{\text{knot}} \cdot a_{\text{DPM}} / (E_{\text{vac, ISM}} \cdot c)$
8	$a_{\text{Aether, freq}}$	$10 \cdot f_{\text{af}} \cdot E_{\text{vac, neb}} \cdot V_{\text{sys}} \cdot a_{\text{DPM}} / (E_{\text{vac, ISM}} \cdot c)$
9	a_{fluid}	$f_{\text{fluid}} \cdot \rho_{\text{ISM}} \cdot V_{\text{sys}} \cdot a_{\text{DPM}} / M$
10	a_{osc}	$2A \cos(kx) \cos(\omega t) + (2\pi/13.8) A \cdot \text{Re}[e^{i(kx - \omega t)}]$
11	a_{exp}	$10 \cdot f_{\text{exp}} \cdot E_{\text{vac, neb}} \cdot V_{\text{sys}} \cdot a_{\text{DPM}} / (E_{\text{vac, ISM}} \cdot c)$
12	f_{TRZ}	0.1 (TRZ constant)
13	(reserved)	—
14	a_{worm}	$f_{\text{worm}} \cdot E_{\text{vac, neb}} / (b^2 + r^2)$

2.2 Wormhole Term (14th)

$$\boxed{a_{\text{worm}}} = \frac{f_{\text{worm}} \cdot E_{\text{vac, neb}}}{b^2 + r^2}$$

where:

$f_{\text{worm}} = 1.0$ — wormhole coupling factor

$E_{\text{vac, neb}} = 7.09 \times 10^{-36} \text{ J/m}^3$ — nebular vacuum energy

$b = 1.0 \text{ m}$ — wormhole throat radius

r = evaluation radius (m)

3. Key Parameters

Symbol	Value	Notes
$E_{\text{vac,neb}}$	$7.09 \times 10^{-36} \text{ J/m}^3$	Canonical (all sessions)
$E_{\text{vac,ISM}}$	$7.09 \times 10^{-37} \text{ J/m}^3$	ISM: $E_{\text{vac,neb}}/10$
f_{DPM}	10^{12} Hz	THz DPM frequency
f_{THz}	10^{12} Hz	THz field
f_{TRZ}	0.1	TRZ constant (Term 12/13)
f_{worm}	1.0	Wormhole factor (Term 14)
b	1.0 m	Wormhole throat
A_{sc}	6.994×10^{18} (or 10^{21})	Cooper super-seeding

4. Wormhole as 14th Term: Physical Justification

4.1 Distinct from PAPER_395

Feature	PAPER_395	PAPER_408
Context	Standalone a_{worm} formula derivation	a_{worm} as 14th additive term in full resonance MUGE
Formula	$a_{\text{worm}} = f_{\text{worm}} \cdot E_{\text{vac}} / (b^2 + r^2)$	Same formula within a 14-term co-sum
Physical role	Independent wormhole acceleration	Resonance MUGE vacuum correction
Code location	<code>compute_a_wormhole()</code>	<code>compute_resonance_MUGE()</code> return sum

4.2 Magnitude Comparison at $r = 10^4 \text{ m}$

$$a_{\text{worm}}(r=10^4) = \frac{1.0 \times 7.09 \times 10^{-36}}{1.0 + (10^4)^2} = \frac{7.09 \times 10^{-36}}{10^8} = 7.09 \times 10^{-44} \text{ m/s}^2$$

Compared to the full 13-term resonance MUGE for SGR1745 ($\sim 1.655 \times 10^{45} \text{ m/s}^2$), the wormhole term at compact scale ($r = 10^4 \text{ m}$) is $\sim 4 \times 10^{-89}$ of the total — **deeply sub-dominant at compact scales** but potentially significant at:

$$r_{\text{cross}} = \sqrt{f_{\text{worm}} \cdot E_{\text{vac,neb}} / a_{\text{DPM}}} - b^2$$

4.3 Large- r Behavior

As $r \rightarrow \infty$: $a_{\text{worm}} \rightarrow 0$ (wormhole decouples from gravity) As $r \rightarrow b$: $a_{\text{worm}} \rightarrow f_{\text{worm}} \cdot E_{\text{vac,neb}} / (2b^2) \approx 3.545 \times 10^{-36} \text{ m/s}^2$

The wormhole term acts as a **near-throat vacuum acceleration** — dominant only within $r \lesssim b = 1 \text{ m}$ of the wormhole throat.

4.4 Term Ordering Significance

Adding the wormhole as **Term 14** (after `f_{\text{TRZ}}` as Term 12/13) follows the construction-file code flow:

```
return aDPM + aTHz + avac_diff + asuper + aaether_res + Ug4i
+ aquantum_freq + aAether_freq + afluid_freq + Osc_term
+ aexp_freq + fTRZ + a_worm;
```

The `// Add wormhole term to resonance MUGE as per updates` comment confirms this is a **deliberate additive extension** of the 12-term formula.

5. Prior 12-Term vs New 14-Term Architecture

Framework	Terms	Reference
PAPER_371	12-term MUGE Superconductive Resonance	Session 101
groksharecfdcad2f5.txt	13-term (adds <code>f_{\text{TRZ}}</code> explicitly as 12th)	Session 107
PAPER_408	14-term (adds <code>a_{\text{worm}}</code> as final term)	Session 108

6. C++ Source

```
// grok_share_cfdcad2f5.txt construction file
double compute_a_wormhole(double r, double f_worm, double Evac_neb, double b) {
return f_worm * Evac_neb * (1.0 / (b * b + r * r));
}

double compute_resonance_MUGE(const MUGESystem& sys,
const ResonanceParams& params) {
double aDPM = /* DPM term ... */;
double aTHz = /* THz cascade ... */;
double avac_diff = /* vacuum differential ... */;
double asuper = /* Cooper super-seeding ... */;
double aaether_res= /* aether resonance ... */;
double Ug4i = /* vacuum BH coupling ... */;
double aquantum = /* quantum frequency ... */;
double aAether = /* aether frequency ... */;
double afluid = /* fluid density ... */;
double Osc_term = /* standing+traveling wave ... */;
double aexp = /* expansion frequency ... */;
double fTRZ = 0.1;

// Add wormhole term to resonance MUGE as per updates
double a_worm = compute_a_wormhole(params.r, params.f_worm,
params.Evac_neb, params.b);

return aDPM + aTHz + avac_diff + asuper + aaether_res + Ug4i
+ aquantum + aAether + afluid + Osc_term + aexp + fTRZ + a_worm;
}
```

7. Relationship to Prior Papers

Paper	Resonance MUGE Form	Notes
PAPER_371	12-term co-sum	First complete resonance framework
PAPER_375	$a_{\text{worm}} = f_{\text{worm}} \cdot \frac{E_{\text{vac}}}{(b^2 + r^2)}$ coupling	Wormhole in advanced integration
PAPER_395	Standalone wormhole acceleration	13th term in prior description
PAPER_408	14-term resonance MUGE with a_{worm} as Term 14	FIRST 14-term complete sum

Whitepaper generated Session 108. Source: groksharecfdcad2f5.txt lines 277-1600.